

Testing Exoplanet Detection of PSR J1719-1438 Using Pulsar Timing

Sahil Shah¹

Abstract

Exoplanets orbiting around pulsars are found to be a rare phenomenon, not many have been found since the start of exoplanet detection. The project explored the method, used by physicists, to test the difficulty of exoplanet detection via pulsar timing. PSR 1719-1438 is a binary system, consisting of a companion with an orbital period of 2.2 hours. Properties about the host neutron star and its companion were derived and explored to determine the nature of pulsar binary systems. The primary objective was to re-obtain the period of the companion using just the data collected by Parkes and Lovell telescope and model it with software such as Tempo2 and SciPy. Variations in the time of arrival residuals were modelled, and optimisation fits were used to accurately determine the period. From the experiment, a value of $0.09070627837842128 \pm 0.00000001425450874$ days was obtained, a value roughly 0.77σ away from the true orbital period of the companion, which is well within the accepted value. Carrying on from this the optimal period was used to make some estimates on the companion's minimum mass, combined separation, and limits on the size of the planet. These results demonstrated the difficulty and precision required to find and study planets around pulsars.

¹*School of Physics and Astronomy, University of Birmingham, Birmingham, United Kingdom*

Contents

1	Introduction	1
2	Scientific Background	2
3	Methods	3
3.1	Deriving the Pulsar Parameters	4
	Maximum radius • Spin-down Luminosity • Surface Magnetic Field • Characteristic Age • Dispersion Measure Distance	
3.2	Proving the Existence of the Planet	5
3.3	Optimising for the Orbital Period	6
3.4	Deriving the Planet's Parameters	6
	Combined Semi-Major Axis • Minimum Mass	
4	Results and Discussion	7
4.1	Deriving the Pulsar's Parameters	7
	Maximum Radius • Spin-down Luminosity • Surface Magnetic Field • Characteristic Age • Dispersion Measure Distance	
4.2	Proving the Existence of the Planet	8
4.3	Optimising for the Orbital Period	9

4.4	Deriving the Planet's Parameters	11
	Combined Semi-Major Axis • Minimum Mass	
4.5	Future Ambition	12
5	Conclusion	12
	Acknowledgments	12
	AI Declaration	12

1. Introduction

Born from the remains of large stars going supernova, neutron stars are dense compact stellar objects. Some neutron stars exhibit conditions of rapid rotation with short periods ranging between a few milliseconds to a couple of seconds, so fast that matter is expelled through the magnetic poles.^[1] The off-shift between the axis of rotation and the magnetic poles, causes regular intervals of flashes to appear when directed at an observers' line of sight behaving as a cosmic lighthouse^[2], as shown in Figure 1. These pulsating stars or better known as pulsars were first discovered in 1967^[3], and are useful in astrophysics to study funda-

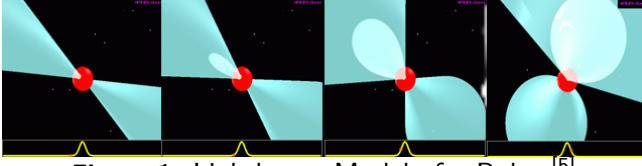


Figure 1. Lighthouse Model of a Pulsar^[5]

mental concepts such as general relativity, gravitational waves, and also discovering exoplanets.^[4]

The discovery of exoplanets orbiting pulsars demonstrates both the precision of pulsar timing techniques and remarkable tenacity of planetary bodies in extreme environments. Since the first detection of the PSR B1257+12 system^[6], pulsar timing has revealed a small but significant number of planetary companions. These systems provide the groundwork for astrophysicists to study planet formation in the remainders of supernovae explosions, behaviour of matter due to strong radiation and physics of compact stellar objects.

This project focuses on PSR J1719-1438, a 5.7 millisecond pulsar in a binary system with a planetary-mass companion^[7]. Its 2.2-hour orbital period companion, likely the remnant core of a carbon-rich white dwarf or evaporated gas giant, experiences one of the most intense environments. This compact binary system raised questions about the structure and timeline of binary millisecond-pulsar systems.

This paper presents a comprehensive pulsar timing analysis of PSR J1719-1438 using data collected over a year in the High Time Resolution Universe Survey for pulsars^[8] using the Parkes and Lovell telescopes. The paper is divided into three principal objectives:

1. Find the companions' orbital period using least-squares fitting of Residuals
2. To derive and characterise fundamental structural properties of the pulsar
3. Find constraints on the system's semi-major axis, planet's mass and radius

The primary objective of the experiment was to test the difficulty of detecting companions using pulsar timing, therefore finding the companion's orbital period was imperative to conduct further research on the system's nature and history. Analysis of the data builds on present day study of compact binary systems, and exoplanet formation. The results not only refined the understanding of the extraordinary system, PSR

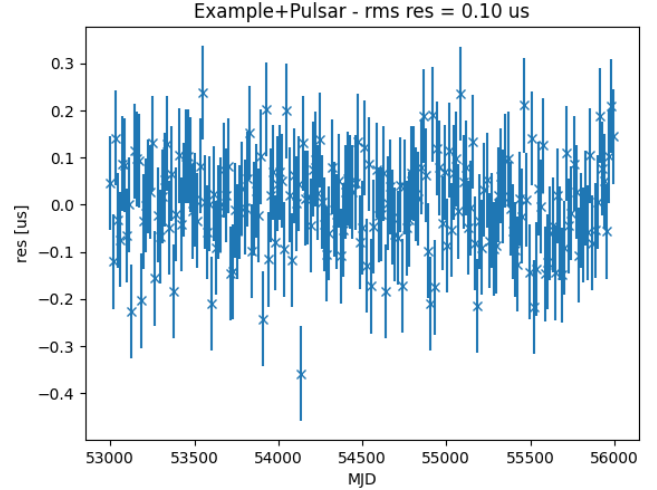


Figure 2. A residual plot of a simulated pulsar, of $P = 15\text{ms}$, as done by a tutorial from Ref.[9]

J1719-1438, but established a clear method for future research in the study of binary pulsar systems.

2. Scientific Background

Pulsar timing relies on the stability of the pulsar's period, millisecond pulsars function as celestial clocks with extreme precision. Following the lighthouse model previously mentioned, shown in Figure 1, the radio waves emitted from the magnetic poles are precisely detected and compared to a timing model. The timing model predicts the time of arrivals (TOAs) by accounting for delays and disturbances between the barycentre of the pulsar system and observer. The approximate timing model is given by:

$$t = t_i - t_0 + \Delta_{\text{clock}} - \Delta_{DM} + \Delta_{R_\odot} + \Delta_{E_\odot} + \Delta_R + \Delta_E + \Delta_S \quad (1)$$

where t_0 is a reference epoch, Δ_{clock} is the clock correction, Δ_{DM} represents the dispersion that occurs through ISM. The remaining terms represent delays in the time taken for the pulse to travel through the pulsar system to our radio telescopes, such as the Shapiro delay Δ_S , which is caused by the additional time taken for light to travel in curves of space-time around heavy objects like the Sun.^[10] The Römer delay^[11], $\Delta_{R_\odot}$, which arises from the travelling time of light across orbits, so if a companion was not accounted for, this term would indicate a strong observable pattern.

Pulsar timing is an extremely precise measurement; therefore, we can measure the deviations between the

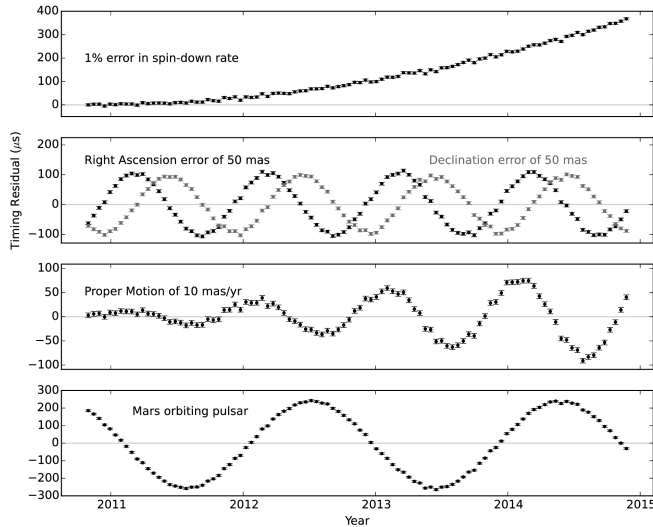


Figure 3. Multiple residual plots for a 3ms pulsar showing how a marginal error can cause the plot to differ largely. [First]: 1% frequency derivative error, [Second]: Right Ascension/Declination error of 50 milliarcseconds(mas), [Third]: Proper motion error of 10 mas/yr, [Fourth]: Mars orbiting pulsar.^[12]

TOA and the model to plot a residual graph which takes the difference of the expected and observed data.

Using the model to create residual-time graphs for pulsars, as shown in Figure 2, which is a simulated pulsar of $P = 15\text{ms}$. A good model would show random points around 0, as the closer to 0 the residual is the better fit the model is to the TOA. Errors on the pulsar's parameters, such as frequency and right ascension, can cause distinct changes in the residual graph, as shown in Figure 3.

From Figure 3, the residual plot of a Mars-like companion creates the strong observable pattern that the Römer delay predicts. The observed pattern can be modelled as a periodic oscillation^[13], producing an effect similar to the Doppler shift for transiting planets. The position of the companion in orbit causes the pulse to arrive either earlier or later than an isolated pulsar. Modelling these periodic variations can reveal information about the planet.

It was through this method; the first set of exoplanets was observed. PSR B1257 + 12 is a pulsar system consisting of 3 exoplanets, found in 1992.^[6] However, despite its success for starting the search for extrasolar planets, pulsar timing is a rare method for detecting exoplanets.^[14] In total only 8 planets have been found using this method, unlike transit method which has been used to find 4360 planets.^[15] The scarcity of de-

tections could arise from the likelihood that exoplanets and pulsars co-exist together. Other challenges could arise from the delays caused by classical and relativistic motion, as well as diminishing factors caused by the interstellar medium (ISM).

3. Methods

For this experiment, the group selected the PSR J1719-1438 pulsar system, a 5.7-millisecond pulsar located 1200 parsec away. First discovered in 2011, using the Parkes 64-meter telescope in a pulsar survey, this system was confirmed to be a binary system.^[7] The companion, PSR J1719-1438 b of mass $1.02M_J$, has an orbital period of approximately 2.2 hours.

This system was chosen from the shortlist of options found in the exoplanet archive as being one consisting of just one companion and one pulsar, which allowed calculations to be fairly easier than systems such as PSR B1620-26 which has a binary star system with a companion.^[16] Moreover, the reason this system was motivated to be studied was the easily accessible data. Found in Ref.[7], Bailes gave supporting documents which provided the parameter and timing file for the pulsar collected using the Parkes radio telescope at frequency of 1368 MHz.

The timing model equation, as shown by equation 1, is a complex equation with multiple factors to consider, therefore Tempo2 was essential to compute the data and provide a model.^[17] Tempo2 is the most convenient software to use when dealing with pulsar timing analysis. The original software is written in FORTRAN and C, and to be able to use this software, a python wrapper known as libstempo^[18] was used, which allows the use of the software using python commands.

Tempo2, takes in two files: the .tim file, which contains the TOAs and the .par file providing the initial parameters of the pulsar. The software then iteratively refines the timing model, using least-squares fitting, in an attempt to reduce the residuals as much as possible. When detecting periodic fluctuations, Tempo2 can estimate binary parameters to correct the model. Therefore, to simulate the timing model for PSR J1719-1438, Tempo2 was used to calculate residuals, then python libraries such as Matplotlib^[19] and SciPy^[20] was used to alter and manipulate the data.

3.1 Deriving the Pulsar Parameters

Further parameters that are not included in the parameter file were determined by using the period and period derivative, assuming that the pulsar behaves as a perfectly spherical rotating body. The period derivative indicated the loss of rotational energy over time, causing the frequency of rotation to decrease. Derivations of the pulsar's parameters were found from Ref.[12], and further research was conducted to find the scientific theory.

From the parameter file, the frequency f and the frequency derivative $\dot{f}$ were given and the frequency was converted to the period taking the reciprocal of the frequency: $P = 1/f$. Using this reciprocity, and taking the derivative with respect to time, the period derivative was obtained:

$$\dot{P} \equiv \frac{d}{dt}(P) = \frac{d}{dt} \frac{1}{f} = -\frac{\dot{f}}{f^2} \quad (2)$$

Using these variables to calculate some derived parameters gives insights about the pulsar's structure and environment. This is fundamental for trying to determine the reason why so few planets have been found orbiting pulsars.

In the case of deriving some of the values, only limits can be placed on the dense object. As a result of this, to avoid complications of limits and true values, an assumption can be made that the pulsar is a conventional pulsar. This indicates a pulsar with $R \approx 10\text{km}$, $M = 1.4M_{\odot}$ and behaves as a uniform density sphere, so it's moment of inertia is $I \approx 10^{38}\text{kg m}^2$. This idea comes the Chandrasekhar limit, which denotes that any white dwarf star of mass $M > 1.4M_{\odot}$ will collapse into a neutron star^[21], therefore assuming all mass is contained during the collapse, the pulsar would have equal mass.

3.1.1 Maximum radius

For a rotating star to remain stable, the centrifugal acceleration that occurs in the equatorial region must not exceed the gravitational acceleration.^[12] Therefore, it must obey the following equality:

$$\omega^2 R < \frac{GM}{R^2}$$

where G is the gravitational constant, M is the mass of the pulsar, R is the radius and $\omega \equiv \frac{2\pi}{P}$. So after rearranging for the radius, the limit on the radius is given by equation 3.

$$R < \left(\frac{GMP^2}{4\pi^2} \right)^{1/3} \quad (3)$$

3.1.2 Spin-down Luminosity

Over time pulsars will lose rotational energy through forms of radiation and friction, causing them to slow down^[1]. The rotational energy, E , can be differentiated to give the rate at which energy is lost, otherwise known as spin-down luminosity.

$$E = \frac{I\omega^2}{2} = \frac{4\pi^2 I}{P^2} \quad (4)$$

The rotational energy, E is given by equation 4, it can assumed that the pulsar conventional and so it is a uniformly dense sphere where its moment of inertia is, $I = 2MR^2/5$.

$$-\frac{d}{dt}E = -\frac{d}{dt} \frac{4\pi^2 I}{P^2} = \frac{4\pi^2 I \dot{P}}{P^3}$$

After differentiating the energy, only the period changes with respect to time, as it can be assumed the mass and radius of the pulsar do not change over the lifetime of the pulsar.

$$-\dot{E} = \frac{4\pi^2 I \dot{P}}{P^3} \quad (5)$$

The spin-down luminosity^[22] is expressed in Watts (or erg s^{-1} if using conventional CGS), as it is energy lost per unit time.

3.1.3 Surface Magnetic Field

If it is assumed that the energy lost is due to the power radiated by the pulsar, it is possible to estimate the minimum magnetic field.^[22] The pulsar can be modelled as a large magnetic dipole that is rotating at some angle θ from the axis of rotation.^[23] The power radiated can be given by

$$P_{\text{rad}} = \frac{2}{3c^3} (BR^3 \sin \theta)^2 \left(\frac{2\pi}{P} \right)^4 \quad (6)$$

where B is the surface magnetic field, c is the speed of light. Equating this to the spin-down luminosity, equation 5:

$$\frac{2}{3c^3} (BR^3 \sin \theta)^2 \left(\frac{2\pi}{P} \right)^4 = \frac{4\pi^2 I \dot{P}}{P^3}$$

$$B^2 = \frac{3c^3 I P \dot{P}}{8\pi^2 R^6 \sin^2 \theta}$$

Assuming that $\theta > 0$, as $\theta \rightarrow 90$, $\sin^2 \theta \rightarrow 1$ and since $B \propto 1/\sin^2 \theta$ the minimum surface magnetic field can be given by the equation:

$$B > \left(\frac{3c^3 I}{8\pi^2 R^6} \right)^{1/2} (P\dot{P})^{1/2} \quad (7)$$

Furthermore, the primary term is just a constant which can be calculated by substituting I and R for a conventional pulsar, so the constant becomes approximately 3.2×10^{19} . The unit of the magnetic field is typically Gauss rather than Tesla as the magnetic field of pulsar is an extremely large value.

3.1.4 Characteristic Age

When pulsars are initially born, periods can be extremely short but are slowed down over time. So, under the assumption that the initial period of the pulsar is much shorter than the observed one, the characteristic age of the pulsar can be calculated^[23]. Although it is not an exact measurement, it is quite precise in measuring the age of the pulsar.

To do this relationship $P\dot{P} = P\dot{P}$ was used and it became $PdP = P\dot{P}dt$, then integrating this, for the limits of the initial and current period for the first integral and 0 to the age τ for the second:

$$\int_{P_0}^P PdP = \int_0^\tau P\dot{P}dt \Rightarrow \frac{P^2 - P_0^2}{2} = P\dot{P}\tau$$

Then using the assumption mentioned before, $P_0 < P$ it was safely said that $P_0^2 \ll P^2$, so P dominates the left hand side.

$$\frac{P^2 - P_0^2}{2} \approx \frac{P^2}{2} = P\dot{P}\tau$$

Rearranging this for τ , the characteristic age of the pulsar is:

$$\tau = \frac{P}{2\dot{P}(365 \times 24 \times 3600)} \text{ years} \quad (8)$$

where units have been converted to years as a better convention for the large time-scale.

3.1.5 Dispersion Measure Distance

Lastly, dispersion measure can be used to find the distance to the pulsars. The dispersion measure is an integrated column density which represents the interactions of radio waves and electrons along the

observer's line of sight.^[24] The interaction causes lower-frequency waves to travel slower than higher-frequency waves as they interact with ISM. The integral:

$$DM = \int_0^d n_e dl = dn_e \text{ pc cm}^{-3}$$

where n_e = the electron number density, d is the distance to the pulsar, represents DM, the dispersion measure. From the parameter file, the dispersion measure is given, typically without errors. Another assumption made is to assume that the pulsar lies in the plane of the galactic disk, such that the number density of electrons $n_e \approx 0.03 \text{ cm}^{-3}$.^[25] Therefore, the distance to the pulsar can be given by equation 9.

$$\text{Distance} = \frac{DM}{0.03 \times 10^3} \text{ pc} \quad (9)$$

3.2 Proving the Existence of the Planet

Before beginning to investigate methods to obtain the period of the planet using optimisation fits, finding evidence to show that there was a periodic variation in the timing model indicating that the companion existed was crucial. From observations, it is unclear that the planet is affecting the pulse's TOA; this is due to the fact that the timing model actually considers the binary parameters and has created a model based on that. So, to begin with, the binary parameters found in the par file were removed, this included the orbital period, semi-major axis, and a setting for Tempo2 to recognise the binary parameters.

The first step in detecting the planet's signal was to apply a phase-fold transformation, equation 10, to all data points, a transformation common to finding orbits in the light curve for the transit method. The function takes all the points and plots them as a fraction of one orbital phase. The phase, ϕ , for each point is given by:

$$\phi = \frac{t - t_o}{P_b} \mod 1 \quad (10)$$

where t is the data point, t_o is a reference beginning time, and P_b is the orbital period of the planet. By plotting the data, we obtained a visible oscillating pattern.

After this, the analysis showed a systematic error in the timing residuals. The model consistently predicted the timing of the pulse wrong, specifically which pulse the timing was measuring. Some of the pulses would

arrive early for the current reading, but the model interpreted it as arriving late for the last pulse. As a solution to this timing discrepancy, a wrap-around function was applied which took the data points that the model was interpreting wrong and added the period of the pulsar, correcting the lateness of the pulse. The final output of this indicated one complete orbital phase plotted against the residuals.

3.3 Optimising for the Orbital Period

The primary aim of the experiment was to obtain this value of the orbital period of the planet using the residual data. Proving the planet existed by implementing the phase fold function used the value of the period, so to work backward, a least-squares minimisation was used to accurately determine the period. The least-squares fitting systematically checked a series of periods within a range to identify which one best fit the observed timing model.

To begin, the model took a short range to iterate through, this was determined from looking at the plot of residual stripped of the binary parameters. For each period, the TOA underwent a phase-fold transformation, given by equation 10, as well as a correction to the timing model to align the data. The data was then fitted to a sinusoidal wave of equation:

$$x(t) = A \sin\left(\frac{2\pi}{P_b}t + C\right) + D \quad (11)$$

where A is the amplitude, P_b is the period, C is the phase offset and D is the vertical offset, using SciPy optimisation. SciPy optimisation used non-linear least-squared minimisation^[20] to create a model, it took the timing model data that was corrected and phase-folded to the test period and fit equation 11 to it, provided the best values of the 4 parameters. The quality of each fit was assessed through the calculation of the chi-squared value.^{[26][27]} Chi-squared is given by the following:

$$\chi_k^2 = \sum_{i=1}^N \frac{(t_i - x(t_i))^2}{\sigma_i^2} \quad (12)$$

where t_i is the observed data point, $x(t_i)$ is the model value, σ_i is the error on the observed data point and k is the degree of freedom. The degree of freedom is determined by the number of constraints on the system, from equation 11, the model is limited to four. The lower the chi-squared value, the better the model

fits, so to minimise this value, the iterative process is run to determine the best possible P_b for the lowest χ^2 . The chi-squared is then reduced using the degree of freedom as a result the closer the reduced chi-squared is to 1 the better the model.

$$\chi_v^2 = \frac{\chi}{v}, \text{ where } v = N - k \quad (13)$$

The least-squares minimisation ended when the model could no longer reduce the chi-squared value any more. From further analysis it was shown, through a residual sum plot, that the optimal orbital period corresponds to the global minimum of harmonic wave. The harmonic wave was produced as a result of the model working for $\pm P_b/2n$. From this, the parameters from the equation 11 were stored to be used as initial guesses for another optimisation. One last optimisation was run with the optimal solutions using the scipy optimise function for the full data set unfolded. The output of this gave a final sine wave with values for all four parameters, that oscillates through the full year of observations.

3.4 Deriving the Planet's Parameters

Once obtaining the orbital period, P_b , of the planet around the pulsar, several fundamental properties can be derived and estimated using orbital mechanics. Using the principles of Keplerian motion of gravitation, the period can determine the semi-major axis, the minimum mass, and the radius.

3.4.1 Combined Semi-Major Axis

The orbital period can be substituted directly into Kepler's third law, which relates the combined separation $a = a_p + a_c$, between the two objects and the orbital period as $a^3 \propto P^2$. Assuming that the mass of the pulsar is much larger than the companion the semi-major axis can be given by:

$$a = \left(\frac{Gm_p P_b^2}{4\pi^2}\right)^{1/3} \quad (14)$$

where m_p is the mass of the pulsar. This value is key for investigating the system and planet formation as the distance between the two stellar objects provide insights behind the dynamical history in the extreme environments.

3.4.2 Minimum Mass

A key characteristic for researching exoplanets is their mass; unfortunately, this requires a lot of measurable

quantities such as the mass of the pulsar and the orbital inclination. Pulsar masses are difficult to accurately measure; only a few have been confirmed, so to solve this, the conventional pulsar can be used. The other limiting factor is the orbital inclination of the system; until this is found, only limits can be placed on the system.

Removing the assumption in Kepler's equation (equation 14) that $m_p \gg m_c$, where m_c is the mass of the companion, the mass function is obtainable. Kepler's third law becomes:

$$P_b^2 = \frac{4\pi^2 a^3}{G(m_p + m_c)} \quad (15)$$

The semi-major axis of the pulsar, a_p , to the common centre of orbit, better known as the barycentre, can be determined using the following equation:

$$a_p = \frac{m_c}{m_p + m_c} a \quad (16)$$

However, this assumes a edge-on orbit, which is uncertain, therefore a $\sin i$ term is added to introduce the angle of inclination i . So to equation 16 this term is added:

$$a_p \sin i = \frac{m_c \sin i}{m_p + m_c} a \rightarrow a = \frac{m_p + m_c}{m_c \sin i} a_p \sin i$$

So, substituting this in 15 and rearranging this to separate variable for terms with mass on the right and measurable on the left, a mass function for the mass of the companion is obtained, equation 17.^[7]

$$f(m_c) = \frac{4\pi^2 (a_p \sin i)^3}{G P_b^2} = \frac{(m_c \sin i)^3}{(m_p + m_c)^2} \quad (17)$$

The pulsar's semi-major axis inclination, $a_p \sin i$, can be measured and is given in the .par file of the pulsar. The masses of pulsars range from around $1.4M_\odot$ to $2M_\odot$, so taking the lower limit (conventional pulsar) and assuming an edge-on orbit ($\sin i = 1$). The minimum mass of the companion can be determined.

4. Results and Discussion

In Ref.[7], Bailes provided the .par and .tim files his team and him used to discover the planet. Using Tempo2 and the raw data provided, Figure 4 shows the residuals for the data collected for PSR J1719 - 1438 over time. The time for this plot is recorded in

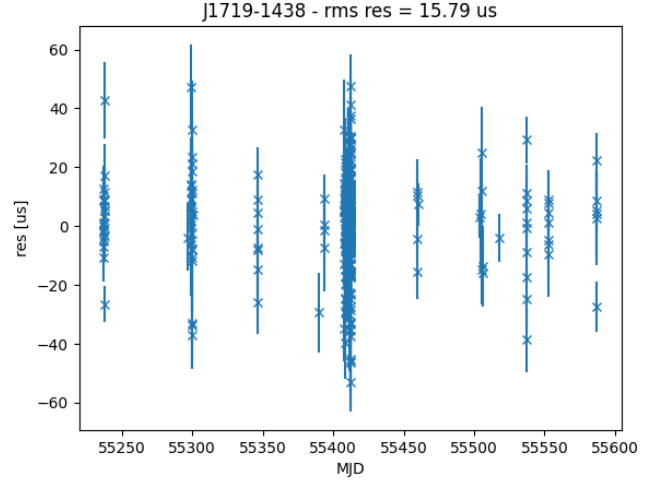


Figure 4. Residual Plot of PSR J1719 - 1438

days, more specifically in modified Julian dates (MJD). The modified Julian dates mark the number of days since midnight on November 17th 1858.^[28]

The timing residual for PSR J1719-1438 has a rms value of $15.92\mu\text{s}$ over a three- and a half-year observing period. This is a minimal deviation from the expected timing model, perhaps suggesting that the model forgot to account for some timing noise or instrumental effects. A solution to this deviation could be further refinement of the ephemeris found in the .par file.

4.1 Deriving the Pulsar's Parameters

Before beginning the main objective of this project, obtaining some of the pulsar's parameters were useful to identify the environment and structure of the pulsar. From Ref.[7], Bailes included a table of ephemeris which included a table of a few derived values. To obtain these characteristics, the period and period derivative can be used. The period of the pulsar, which was obtained using the reciprocal of the frequency, was $P = 0.0057902(\pm 6.3996 \times 10^{-15})\text{s}$, which has a relative error $\sigma_P/P \ll 1\%$, indicating how precisely it was measured. Using equation 2, the period derivative is $\dot{P} = (7.3619 \pm 0.4576) \times 10^{-21}$, which is the time increase of the period per second. Using the following two values, the remaining were derived.

$$\sigma_F^2 = \left(\frac{\partial F}{\partial A} \sigma_A\right)^2 + \left(\frac{\partial F}{\partial B} \sigma_B\right)^2 + \left(\frac{\partial F}{\partial C} \sigma_C\right)^2 + \dots \quad (18)$$

For the errors on the values, these were calculated using the error propagation formula, given by equation 18, where $F = F(A, B, C, \dots)$. For example, the error

on the period derivative can be calculated using this formula where $F(A, B, C, \dots) = \dot{P}(f, \dot{f})$. Computing this:

$$\sigma_P^2 = \left(\frac{\partial \dot{P}}{\partial f} \sigma_f\right)^2 + \left(\frac{\partial \dot{P}}{\partial \dot{f}} \sigma_{\dot{f}}\right)^2$$

$$\sigma_P^2 = \left(\frac{2\dot{f}}{f^3} \sigma_f\right)^2 + \left(-\frac{1}{f^2} \sigma_{\dot{f}}\right)^2$$

While the statistical distances between the derived values and the literature values was given by:

$$\sigma = \frac{|x_1 - x_2|}{\sqrt{\sigma_1^2 + \sigma_2^2}} \quad (19)$$

where the $_1$ depicts the value obtained and $_2$ represents literature value.

4.1.1 Maximum Radius

Obtaining a maximum radius for the pulsar was an additional parameter that was thought to be useful. It describes the structure of a stable rotating body's limit. Using equation 3 and using the period of the pulsar and the conventional pulsar mass of $1.4M_\odot$, the limit of the radius of PSR J1719-1438 was $R < 54.03643 \pm 0.00057 \text{ km}$. This value is considered within a normal range; the idea of a conventional pulsar falls into this equality. The true value of the radius is not determined, but this range provides useful information, for this case of uncertainty the conventional radius of a neutron star was used.

4.1.2 Spin-down Luminosity

An important value for pulsars is the energy they lose over time; this value provides information about the energy radiated toward the companion. For PSR J1719-1438, given by equation 4, the rotational energy indicated that the pulsar has approximately $(6.55607 \pm 0.00015) \times 10^{43} \text{ Joules}$, which is equivalent for the Sun to be powered for billion years. This immense storage of energy is dissipated through radiation and braking force, which could influence both the evolution of the pulsar and its companion.

Upon calculating the spin down luminosity from equation 5, the pulsar radiates $(0.44 \pm 0.03)L_\odot$, where $L_\odot$ is the luminosity of the Sun. This value is consistent with the value of $(0.40 \pm 0.04)L_\odot$, provided in Ref.[7] to within 0.74 standard deviations, which shows excellent consistency. The small discrepancy suggests some error caused by uncertainties in the calculations, perhaps due to assumptions made about the conventional pulsar.

4.1.3 Surface Magnetic Field

The surface magnetic field of millisecond pulsars provides information about the evolution and current electromagnetic properties. From the use of equation 7, the magnetic field of PSR 1719-1438 was contained to $B_s > (2.10 \pm 0.07) \times 10^8 \text{ G}$, while measures from Ref.[7] reported a maximum limit of $B_s < 2 \times 10^8 \text{ G}$. The derived lower bound exceeds the maximum limit in the literature, and the absolute limit of the two values has a difference of 1.4σ . According to the ANTF pulsar catalogue^[29], $B_s = 2.18 \times 10^8 \text{ G}$, which is consistent with the derived range. Normal magnetic field of pulsars are much higher than 10^8 G with approximate ranges of $10^{12} - 10^{15} \text{ G}$ ^[30], suggesting that this pulsar has something odd with its evolution.

4.1.4 Characteristic Age

The characteristic age of pulsars indicates the evolutionary path and timescale for compact stellar objects, as well as limits to place on the planet formation timeline. The analysis indicated that the pulsar was at least $\tau > (12.5 \pm 0.8) \text{ Gyr}$, similar to the paper^[7], producing a value of $\tau > 12.5 \text{ Gyr}$. This shows excellent agreement with 0 statistical differences. This result confirms that the pulsar is very old, being only a billion years younger than the age of the universe. This age contradicts the actual period of the pulsar, as older pulsar slow down but this is still a millisecond pulsar. However, this agrees with the lower magnetic field suggesting the pulsar is actually recycled pulsar.^[31]

Recycled pulsars occur when stars are born in a binary together, and when one goes supernova if the remaining star is not ejected, the pulsar will eventually start absorbing the matter around the end-cycle star. This causes the pulsar to start spinning faster and reduce its magnetic field.^{[7][31]} This model seems probable to the pulsar's characteristics.

4.1.5 Dispersion Measure Distance

The derived distance to the pulsar using the dispersion-distance relation produced a value of $1.23 \pm 0.41 \text{ kpc}$ which demonstrates near perfect agreement with the established value^[7] of $1.20 \pm 0.30 \text{ kpc}$, differing by 0.05σ , this value suggests that the pulsar lies well within the galactic disk of the galaxy, Once again supporting the idea of being a recycle pulsar.

4.2 Proving the Existence of the Planet

The residual plot, Figure 4, did not indicate a clear variation in the timings. The plot has thick bands of

data showing the days that the telescope was observing but no period variations were identified. Once the binary parameters were removed, the data started to show more trends as shown in Figure 5a. Two key patterns can be seen from this figure, the first being the thick band of data points recorded on MJD 55412 indicated by the vertical red lines. Secondly, there was a systematic error occurring throughout the observed data, which were the points below -0.002 .

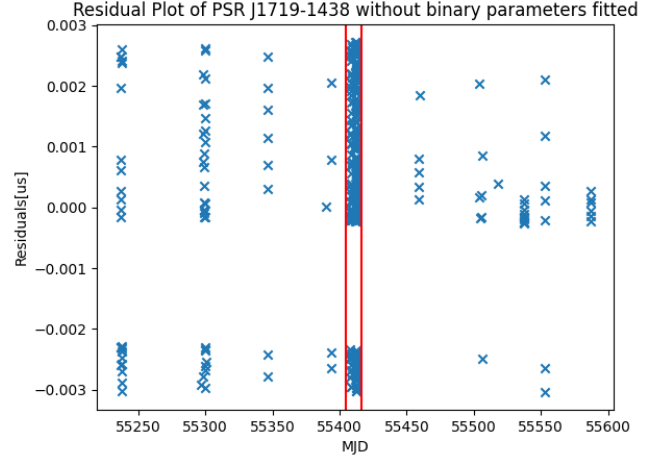
Analysing the thick band of data by zooming in, shown in Figure 5b, a clear dip can be seen, which resembles a sinusoidal pattern. From further deduction, looking at the x-axis it is clear that the approximate period of this wave, considering the systematic error, is around 0.09 days, which is also the orbital period of the planet. If this were to be an orbital period, the rest of the data points would have to follow the same pattern, so applying the phase fold function given by equation 10 to all the data points, using the orbital period given in the parameter file, the data points folded very consistently as shown in Figure 6a. Here, the systematic error computed by the timing model is very evident, occurring at the maxima of the wave. The peak of the wave has been shifted downward, so to correct all points below -0.002 a correction was made where the period of the pulsar was added.

Ultimately, from Figure 6b, it is clear that there was a planet orbiting the pulsar that caused these periodic fluctuations to the TOA and timing model.

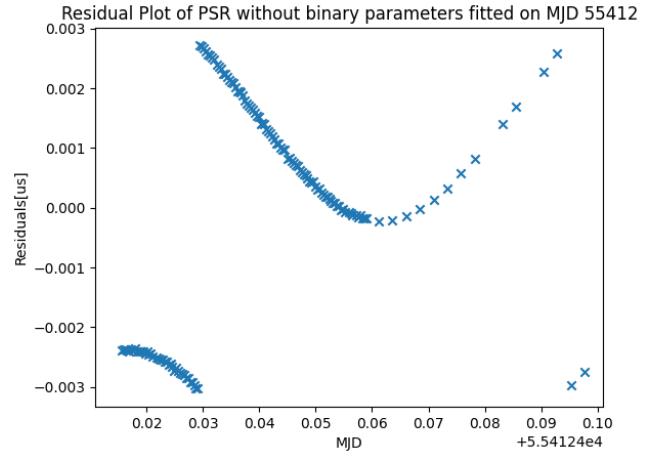
4.3 Optimising for the Orbital Period

A preliminary attempt to optimise the orbital period was made on the full data set corrected for the timing error; however, this attempt was not successful. The primary issue the SciPy optimisation function ran into was the blank space between the observations. The model struggled to fit a complete sine wave to the incomplete data set. As a result, a secondary fit was performed on the data points where the periodic wave could be seen, as shown in Figure 5b. To complete this fit, the .tim file was filtered to show just the data set for the 4th of August (MJD 55412), this was then run through the Tempo2 software and a similar timing correction had to be applied. The timing model inverted the systematic error, so the correction was now for $t > 0.002$ and P was subtracted to these data points.

Now applying the SciPy optimisation function to fit a curve with equation 11 to this data set returned an es-



(a) Residual plot when the binary parameters removed from the .par file

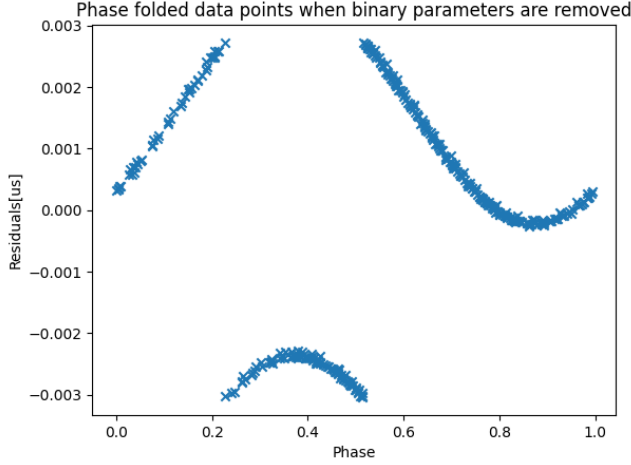


(b) Residual plot for MJD 55412, indicating a periodic wave

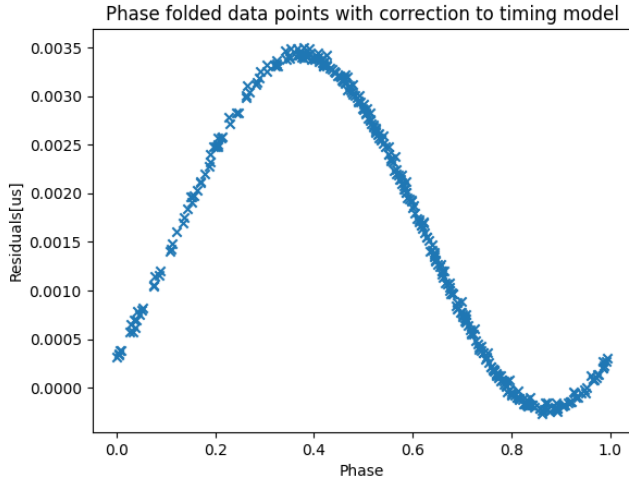
Figure 5. All data points manipulated to find one complete orbital phase of the planet

timate of $P_b = 0.0909998243698 \pm 0.0001484493486$ days. The sine wave was then plotted to the reduced data set, as shown in Figure 7, this fit accurately represented this data. This value is approximately 1.98σ away from the true value, this statistical distance is not very large, but this value is not accurate. The estimate of this value was only dependent on the single day of observations rather than the 1-year observational period.

The phase-folded graph (Figure 6b) was a great form of representing the orbital motion and was dependent on the period. Therefore, to obtain the period, an iterative process of least squares was run to find the best value. The process compared a test pe-



(a) Residual plot when phase folding function is applied



(b) Phase Folded data once correction to timing model is applied

Figure 6. All data points manipulated to find one complete orbital phase of the planet

riod that folded the data and produced a sine wave with a sine wave that should have a period 1. The model calculated how close the test model was to the true model using the chi-squared value (equation 12), and compared it with the previous test, the lower chi-squared was kept. The optimal period produced by this least squares process returned a value of $P_b = 0.0907062969999931$, producing a phase-folded graph nearly identical to the phase-folded graph in Figure 6b. This model had the potential to produce a more precise value; however, due to timing constraints in the iterative process, the lowest iterative step taken was 1×10^{-9} . Decreasing the step length by a factor of 10 would multiply the processing time by 10, which

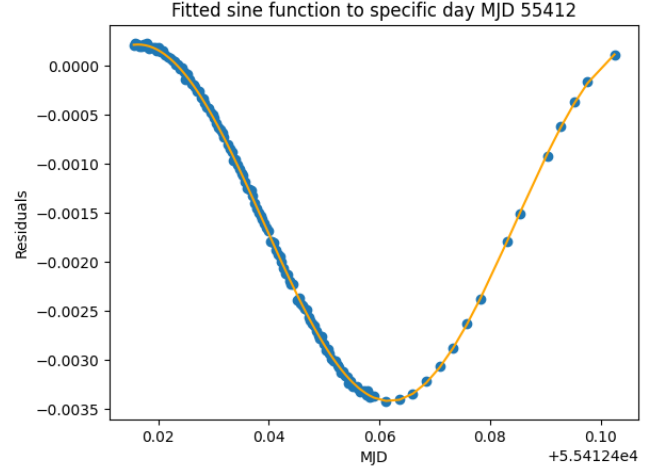


Figure 7. SciPy Optimised sine wave for just MJD 55412

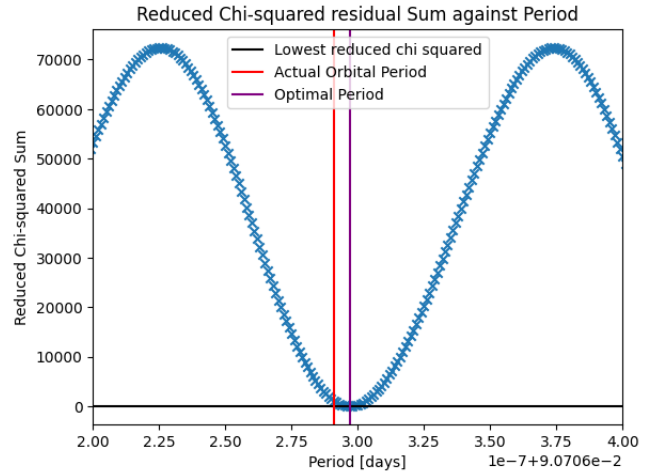
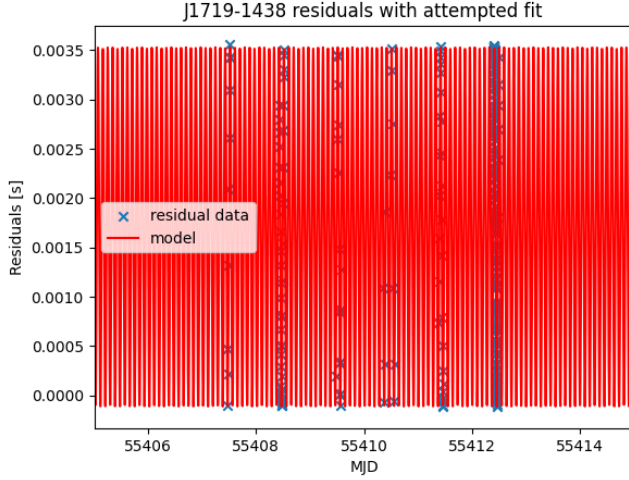


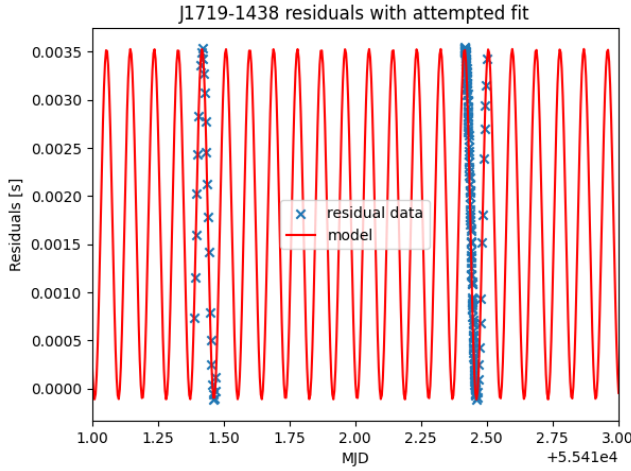
Figure 8. Reduced chi-squared sum against period, the global minimum indicating the optimal period

was a limitation that was compromised on.

When computing the reduced chi-squared value for optimal, using equation 13 with 4 degrees of freedom, the lowest value obtained was $\chi^2_4 = 2.2831$, which suggests that the model has been under-fitting the data. From Figure 8, the plot of the residual sum compared to the period, the global minimum represents the optimal period (purple) for the minimised reduced chi-squared value (black). The sharpness of the minima curve suggests a well-constrained orbital period and compared with the actual orbital period (red) the value does not overlap with the optimal period. This could be a consequence of assuming a Keplerian orbit. Periodic waves occur when the body orbits in circular motion, not accounting for imperfect orbits (elliptical) with eccentricity. Therefore, the difference in the



(a) Model of sine wave using the final period for all data [Region: MJD 55405 - 55415]



(b) Better visual of sine wave for MJD 5511-55413 fitted against the residual data

Figure 9. Final model of periodic wave to simulate periodic oscillation of residual data

two values for the optimisation of a sine wave occurs because the true motion is not circular.

After obtaining the optimal period, another SciPy optimisation was performed using the optimal sine wave parameters as the initial guesses in the curve fit function. This final optimisation gave a result of $P_b = 0.090706278378 \pm 0.000000014255$ days. The final result agrees much better than the initial optimisation, within the range of 0.77σ from the true period. The plot of this sine wave to indicate the orbital motion of the planet can be seen in Figure 9a. Hard to determine whether this model is a good fit due to the large density of transits per day, a zoomed-in plot, shown by Figure 9b, between days MJD 55411 and 55413 provides a

clear visual for the accuracy of the model with the data. The model precisely crosses all data points, providing further motion during the absence of observed data.

4.4 Deriving the Planet's Parameters

Following the optimised value of the period, the companion's parameters were now calculated using Keplerian motion.

4.4.1 Combined Semi-Major Axis

Using equation 14, where P_b is now the optimised period and m_p is the mass of a conventional pulsar of $1.4M_\odot$, the approximate orbital radius of the companion is $661191667.17 \pm 7005.32\text{m}$ which when converted to astronomical units is $(4.41979 \pm 0.00005) \times 10^{-3} \text{ au}$. This makes it one of the most compact radio pulsar binaries to exist^[7], which gives rise to theories about the formation of this planet. A planet forming that close to its host star must have special properties and tests whether the planet was formed before or after the formation of the pulsar. The calculated value was not mentioned in the paper, but according to Ref.[15] the orbital radius is 0.0044au ; which is identical to the value derived.

4.4.2 Minimum Mass

Using the mass function, given by equation 17, the minimum mass of the pulsar was calculated. Beginning with the left-hand side of the function, which is independent of the mass, the constant $a_p \sin i$ was given in the ephemeris table in Ref.[7] with a value of 0.001819 ± 0.000001 light seconds. Once converted metres, the function of the mass companion was $f(m_c) = (1.5618 \pm .0026) \times 10^{21} \text{ kg}$. Moving on to the right-hand side, which was dependent on both m_c and i , rearranging the equation and once again assuming that $m_p \gg m_c$ it became:

$$m_c \sin i = (m_p^2 f(m_c))^{1/3}$$

where $0 < i \leq 90^\circ$. Shifting the $\sin i$ to the right side made $m_c \propto (\sin i)^{-1}$, therefore when $\sin i = 1$, m_c is minimised and as $i \rightarrow 0$, m_c increases. So, assuming an edge-on orbit, the minimum mass of the companion was $m_c > 2.296 \pm 0.001 \times 10^{27} \text{ kg}$, which was converted to $m_c > (1.155 \pm 0.001) \times 10^{-3} M_\odot$. This value matches the literature value provided in Ref.[7]. The planet's mass is then compared to Jupiter's mass, the planet has a mass of equal magnitude to Jupiter, which was determined as $m_c \geq 1.210 \pm 0.001 M_J$, where M_J is the mass of Jupiter, still matching literature values.^[15]

4.5 Future Ambition

Due to timing constraints, a value that was not derived which would have been useful to know was the radius of the planet. In Ref.[7], the combined orbital separation as well as the Roche-Lobe limit^[32] was combined to give the so-called mean density-orbital relation^[33], a lower limit on ρ for the planet. Using the orbital period, this would've provided insight about the structure of the planet, a crucial vision for determining the type of system this is. Through this, it was discovered that the density of the companion $\rho > 23\text{g cm}^{-3}$, this value indicates that the companion is not an average planet but perhaps a remnant of a star, as typical gas giant like Jupiter have a density of $\rho < 2\text{g cm}^{-3}$, which agrees with the idea of PSR J1719-148 being a recycled pulsar^[7].

5. Conclusion

This experiment successfully determined that the orbital period of PSR J1719-1438 b was $0.090706278 \pm 0.000000014$ days by optimisation of the least squares of the phase-fold function. The period was accurately supported through the global minimum of the reduced χ^2 goodness of fit, although there was some discrepancy between the actual value and the optimal period. The derived parameters of the pulsar, such as the spin-down luminosity ($-\dot{E} = 0.44L_{\odot}$), surface magnetic flux ($B_s \approx 2.1 \times 10^8$ G), characteristic age ($\tau \approx 12.5$) GYr, and distance (1.2 kpc) were all met and accepted and did not produce anything greater than 2σ from literature values. This provided insight into the nature and structure of pulsars as well as the environment surrounding the pulsars that the planet experiences. The pulsar follows a model of being recycled by the companion.

Following the re-obtained value of the orbital period, which was accurate to 0.77σ of the true orbital period, the calculations for the planet's parameters did not differ from literature values very largely. Obtaining the combined separation of the binary system to be $a = 0.0044$ au and the minimum mass of the companion to be roughly larger than that of Jupiter at $1.20M_J$. Further analysis would've showed consistency with the companion forming before the supernova explosion and causing the pulsar to speed up through the process of recycling.

This study effectively demonstrated the precision of finding exoplanets in binary systems around pulsars

clarifying why the use of pulsar timing is an uncommon method for detecting planetary systems. Future studies using better radio telescopes such as the planned Square Kilometre Array^[34] could further refine the data, providing information about the composition, and emissions. As well as accounting for further errors in the model such as delays and ISM dispersion could resolve further discrepancies. These steps would further describe the evolutionary history of not just PSR J1719-1438 but millisecond pulsar binaries in general.

Acknowledgments

Henry Bedford - Conducted initial research on pulsar timing for exoplanets, worked on methods for proving the existence of the planet (section 4.2), conducted initial optimisation for one day (section 4.2), was the main contributor for determining the orbital period, coded the least squares and chi-squared values (section 4.3).

Sam Hollier - Conducted initial research on pulsar timing for exoplanets, worked on methods for proving the existence of the planet (section 4.2), conducted initial optimisation for one day (section 4.2), as well working on helping with (section 4.3) and future research on the Roche-Lobe/density-period relation (section 4.5).

Sahil Shah - Conducted initial research on pulsar timing for exoplanets, worked on methods for proving the existence of the planet (section 4.2), conducted initial optimisation for one day (section 4.2), and completed entirety of pulsars parameter derivations (section 4.1)

Will Thistle - Conducted initial research on pulsar timing for exoplanets, worked on methods for proving the existence of the planet (section 4.2), conducted initial optimisation for one day (section 4.2), and completed entirety of planets parameter derivations (section 4.4)

AI Declaration

AI was not used during the project to help with research or to code.

References

- [1] Thomas Gold. Rotating neutron stars as the origin of the pulsating radio sources. *Nature*, 218(5143):731–732, 1968.
- [2] HO De Oliveira, RM Marinho, and NS Magalhaes. Stellar structures and the enigma of pulsars rotation frequency decay. In *Journal of Physics: Conference Series*, volume 630, page 012059. IOP Publishing, 2015.
- [3] SJ BELL, RA COLLINS, A HEWISH, JDH PILKINGTON, and PF SCOTT. Observation of a rapidly pulsating radio source(pulsating radio source with repetition period of extreme constancy, suggesting white dwarf or neutron star oscillation in galaxy as sources). 1968.
- [4] JM Cordes, M Kramer, TJW Lazio, BW Stappers, DC Backer, and S Johnston. Pulsars as tools for fundamental physics & astrophysics. *New Astronomy Reviews*, 48(11-12):1413–1438, 2004.
- [5] Duncan Lorimer. Binary and millisecond pulsars at the new millennium. *Living Reviews in Relativity*, 4, 06 2001.
- [6] Aleksander Wolszczan and Dail A Frail. A planetary system around the millisecond pulsar psr1257+ 12. *Nature*, 355(6356):145–147, 1992.
- [7] Matthew Bailes, Stuart D Bates, Varun Bhalerao, ND Ramesh Bhat, Marta Burgay, Sarah Burke-Spolaor, Nichi D’Amico, Simon Johnston, Michael J Keith, Michael Kramer, et al. Transformation of a star into a planet in a millisecond pulsar binary. *Science*, 333(6050):1717–1720, 2011.
- [8] MJ Keith, A Jameson, Willem van Straten, Matthew Bailes, S Johnston, M Kramer, A Posenti, SD Bates, NDR Bhat, M Burgay, et al. The high time resolution universe pulsar survey—i. system configuration and initial discoveries. *Monthly Notices of the Royal Astronomical Society*, 409(2):619–627, 2010.
- [9] Libstempo Tutorial of a Simulated Pulsar https://github.com/VIPER-Research-Group/gw-school-2022-materials/blob/master/Week_1/Day_2/timing_tutorial/timing_libstempo_intro.ipynb.
- [10] Irwin I Shapiro. Fourth test of general relativity. *Physical Review Letters*, 13(26):789, 1964.
- [11] M Romer and I Bernard Cohen. Roemer and the first determination of the velocity of light (1676). *Isis*, 31(2):327–379, 1940.
- [12] Scott M. Ransom James J. Condon. Section 6: Pulsars in Essential Radio Astronomy <https://www.cv.nrao.edu/~sransom/web/Ch6.html>.
- [13] EA Behrens, SM Ransom, DR Madison, Z Arzoumanian, K Crowter, ME DeCesar, PB Demorest, T Dolch, JA Ellis, RD Ferdman, et al. The nanograv 11 yr data set: Constraints on planetary masses around 45 millisecond pulsars. *The Astrophysical Journal Letters*, 893(1):L8, 2020.
- [14] Rebecca G Martin, Mario Livio, and Divya Palaniswamy. Why are pulsar planets rare? *The Astrophysical Journal*, 832(2):122, 2016.
- [15] Nasa Exoplanet Archive for Pulsar Timing - https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbls&config=PS&constraint=default_flag>0&constraint=disc_method%20like%20%27%25pul%25%27.
- [16] Steinn Sigurdsson, Harvey B Richer, Brad M Hansen, Ingrid H Stairs, and Stephen E Thorsett. A young white dwarf companion to pulsar b1620-26: evidence for early planet formation. *Science*, 301(5630):193–196, 2003.
- [17] GB Hobbs, RT Edwards, and RN Manchester. Tempo2, a new pulsar-timing package—i. an overview. *Monthly Notices of the Royal Astronomical Society*, 369(2):655–672, 2006.
- [18] Michele Vallisneri. libstempo: Python wrapper for tempo2. *Astrophysics Source Code Library*, pages ascl-2002, 2020.
- [19] J. D. Hunter. Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.
- [20] Ralf Gommers, Pauli Virtanen, Matt Haberland, Evgeni Burovski, Tyler Reddy, Warren Weckesser, Travis E Oliphant, David Cournapeau, Andrew Nelson, Pamphile Roy, et al. *scipy/scipy: Scipy 1.15.0*. Zenodo, 2024.

- [21] Subrahmanyan Chandrasekhar. The maximum mass of ideal white dwarfs. *Astrophysical Journal*, vol. 74, p. 81, 74:81, 1931.
- [22] D.R. Lorimer and M. Kramer. *Handbook of pulsar astronomy*, volume 4. Cambridge university press, 2005.
- [23] Alice K Harding, Ioannis Contopoulos, and Demosthenes Kazanas. Magnetar spin-down. *The Astrophysical Journal*, 525(2):L125, 1999.
- [24] Xiao-Peng You, G Hobbs, William A Coles, Richard N Manchester, R Edwards, Matthew Bailes, J Sarkissian, Joris PW Verbiest, Willem Van Straten, Aidan Hotan, et al. Dispersion measure variations and their effect on precision pulsar timing. *Monthly Notices of the Royal Astronomical Society*, 378(2):493–506, 2007.
- [25] Miguel A de Avillez, Ashish Asgekar, Dieter Breitschwerdt, and Emanuele Spitoni. Electron distribution in the galactic disc: results from a non-equilibrium ionization model of the interstellar medium. *Monthly Notices of the Royal Astronomical Society: Letters*, 423(1):L107–L111, 2012.
- [26] William G. Cochran. The χ^2 test of goodness of fit. *The Annals of Mathematical Statistics*, 23(3):315–345, 1952.
- [27] Karl Pearson. X. on the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 50(302):157–175, 1900.
- [28] Peter Meyer. Julian day numbers. *Hermetic Systems*. http://www.hermetic.ch/cal_stud/jdn.htm, pages 491–499, 2002.
- [29] R N Manchester, G B Hobbs, A Teoh, and M. Hobbs. The australia telescope national facility pulsar catalogue. *The Astronomical Journal*, 129(4):1993, 2005.
- [30] Dipanjan Mukherjee, Peter Bult, Michiel van der Klis, and Dipankar Bhattacharya. The magnetic-field strengths of accreting millisecond pulsars. *Monthly Notices of the Royal Astronomical Society*, 452(4):3994–4012, 2015.
- [31] Dipankar Bhattacharya and Edeard Peter Jacobus van den Heuvel. Formation and evolution of binary and millisecond radio pulsars. *Physics Reports*, 203(1-2):1–124, 1991.
- [32] B Paczynski. Evolutionary processes in close binary systems. *Annual Review of Astronomy and Astrophysics*, vol. 9, p. 183, 9:183, 1971.
- [33] Juhan Frank, Andrew R King, and Derek Raine. *Accretion power in astrophysics*. Cambridge university press, 2002.
- [34] Peter E. Dewdney, Peter J. Hall, Richard T. Schilizzi, and T. Joseph L. W. Lazio. The square kilometre array. *Proceedings of the IEEE*, 97(8):1482–1496, 2009.